

Procedural Living Worlds

A scalable rule engine and an agent-assisted workshop for the Living World Model

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Abstract

A convincing game world should keep moving when the player leaves the room. People should eat, work, look after one another and pursue plans. Animals should respond to their surroundings. Buildings, objects and weather should change in ways that later scenes can explain. Calling a language model for every small action would make a large world expensive and difficult to reproduce.

This report proposes a **Procedural World Layer** for the Living World Model. Compact rules run the ordinary world on CPUs. Storyteller models add detailed interpretation and dramatic development inside selected Simulation Fields. Both produce changes through the same world coordinator and retain the same identities, histories and character perspectives. A separate **Rule Workshop** uses coding agents to propose improvements, replay test worlds, evaluate behavior and submit versioned rule packages for review. Runtime simulation and rule development have different budgets and different permissions.

The design combines readable classes, reusable components, explicit action contracts, probabilistic decisions and an event scheduler. It specifies how to exchange control with Storyteller fields, test improvements without rewarding implausible drama, and scale experiments across many workers. Capacity tables distinguish exact storage arithmetic from assumed processing costs. No simulation benchmark or automatic improvement result is claimed.

The vocabulary in one minute

Term	Meaning in this report
Living World Model (LWM)	The overall memory and coordination architecture for a persistent interactive world
Simulation Field	A selected working set of people, objects, places and events that must be considered together
Simulated agent or actor	A person, animal, robot or other entity capable of acting
Event-Entity	A persistent scene or activity, such as a journey or conversation
Beat	An ordered record of changes over an interval of world time
Development agent	A coding or evaluation worker improving rules outside the running world

How to read this report

Pages 1-13 are the approximately twenty-minute conceptual route: purpose, vocabulary, rules, control handoff, evaluation and scale. Appendices A-E provide implementation contracts, worked calculations and a prototype plan. The existing LWM whitepaper is the companion architecture [1]. Here, “Sims-like” describes the desired experience, not a claim about The Sims’ internal implementation.

1 One world, two ways to advance it

The LWM's **Temporal World Knowledge Graph** is a time-aware factual memory. It connects entities with facts such as “Alice holds the ring until 14:23” and “Devon starts work at 09:00.” Each entity's timeline is a view into that shared history. A character's perceptions, memories and beliefs are separate records: Devon can be wrong about where the ring is without changing its actual location.

The new Procedural World Layer adds a cheap way to advance this memory. Rules handle routines and measurable consequences. A Storyteller handles selected scenes with richer interpretation, dialogue and mental state. The **Attention Allocator** decides what requires attention, how often to update it and how much compute to spend. A Simulation Field can follow a player, stay with an important NPC, or connect distant characters during a call.



Figure 1. Both simulation methods propose Beats to one coordinator. Accepted history feeds the next update. The Rule Workshop publishes tested rule versions; it does not edit a live character's past.

The distinction that makes scale possible

At 08:00, Alice walks toward work while Devon sleeps. Alice may need meter-level navigation and second-level reactions. Devon may need only a calculated wake time and a gradually changing hunger value. Neither requires a fresh paragraph every second. When the player knocks on Devon's door, the coordinator brings his current state forward and transfers the appropriate decisions into a Storyteller field.

Temporal detail, semantic detail and presentation detail remain separate. The LWM's L0-L5 profiles range from regional summaries to relevant embodied experience [1]. Procedural rules can support several profiles. A field is a coordination boundary, not a requirement that every operation inside it uses an LLM. Physics, timers and object damage can stay procedural during a richly narrated scene.

2 Building blocks a designer can understand

Start with a neighborhood: people, animals, homes, a workplace, furniture, paths and changing conditions. A **class** describes a kind of thing. An **instance** is one particular thing. A **component** is a reusable bundle of properties and behavior, such as Needs, Inventory or Durability. An **affordance** is an action an object offers: a chair offers Sit, a guitar offers Play, and a door offers Open.

Building block	Typical properties	Methods or capabilities
Entity	Stable ID, name, kind, creation time	Inspect state; receive a validated change
Actor	Body, needs, goals, skills, beliefs	Perceive; consider actions; attempt one
Human / animal / robot profile	Physiology, senses, movement, developmental rules	Define supported needs and capabilities
Object	Position, footprint, mass, condition, affordances	Offer interaction; receive effects
Location	Boundary, entrances, capacity, schedule, ownership	Test membership; admit or restrict entry
Process	Start time, parameters, affected region	Advance weather, decay, fire or another ongoing change
Rule package	Conditions, effects, dependencies, version	Propose changes and explain their causes

Use inheritance for stable kinds

A Human can inherit shared Actor behavior. A Door can inherit shared Object behavior. This helps authors find familiar defaults. Keep the hierarchy shallow. A character should not become an entirely different class whenever they find a job, move country or have a birthday.

Use components and profiles for changeable facts: life stage, profession, household role, preferences, learned norms and current effects. A nurse who becomes unemployed keeps their skills and relationships; the Employment component changes. A robot may have battery charge and maintenance needs but no digestive needs. A toddler may need a caregiver and have fewer action capabilities. These differences are explicit rather than scattered through every rule.

This authoring model can look object-oriented while its runtime stores numbers in compact arrays. The distinction is like a spreadsheet: designers understand each row as a person, while the engine efficiently processes a whole column of hunger values. Appendix A makes this mapping concrete.

3 What a rule actually says

A useful rule says **when it applies, what information it uses, what it proposes, how long that takes and what changes if it succeeds or fails**. It does not simply order “Alice goes to work.” It asks whether work is due, whether Alice knows that, whether she can travel, and whether another need or commitment takes priority.

At 08:40	What the engine considers	Possible result
Alice is employed; shift starts at 09:00	Travel takes 15 minutes; leaving now has high value	Prepare to leave
Bladder urgency is high	A usable toilet is nearby; using it takes three minutes	Reserve toilet, walk there, use it, then leave
Toilet is occupied	Compare waiting, another toilet and lateness	Queue or choose a feasible alternative
A child needs immediate help	A care obligation interrupts the work plan	Help, then reassess travel and notify work if appropriate
Devon is unemployed and bored	No work deadline; television and guitar are available	Choose according to preferences, effort and context

A decision is different from an action

The engine first gathers feasible actions, then ranks or samples them. A selected action becomes a small plan with duration and progress. It may reserve an object, move to an interaction point, produce effects and eventually finish. If the room catches fire, the plan can be interrupted. The engine does not choose a completely new hobby every fraction of a second.

A rule may be deterministic: a locked door cannot be opened without an allowed method. It may be probabilistic: under one test configuration, Alice chooses the toilet first in 80% of eligible cases. That percentage is a design parameter, not a measured human fact. It applies once to this decision, not repeatedly on every update until the desired outcome appears.

Resolve conflict explicitly

First enforce feasibility and urgent constraints. Then consider commitments, needs, preferences, social expectations and opportunity cost. If two rules want to change the same property, use declared combination rules or one conflict resolver. An agent author cannot silently make their module win by writing an arbitrarily large priority number.

4 Needs create pressure; context shapes the response

A need is a changing state, not a script. Hunger, thirst, bladder urgency, fatigue, comfort, social connection and stimulation can each have a level, a rate of change and thresholds. An adult profile may also include intimacy motives. A species or robot profile selects the needs it actually supports. The simulator must avoid treating all actors as the same human template.

In a simple model, need values range from 0 to 1, where 1 means urgent. They grow with elapsed world time, respond to events and fall through suitable actions. A distressing event may increase one character's wish for company while another prefers solitude. That response should follow an authored individual profile, relationship and prior experience, rather than a universal rule about grief.

```
new_need = clamp(old_need + rate * elapsed_time
                 + event_effects - action_relief, 0, 1)

action_value = expected_need_relief + goal_progress
              + preference_fit + commitment_value
              - time_cost - risk - social_cost
```

These formulas are starting policies, not a complete theory of motivation. Units and scales must be declared. A critical thirst constraint may override the ordinary score. Social opportunity may depend on another person's willingness, so a lonely character cannot obtain conversation by changing the other person's state directly.

Preserve identity without forcing sameness

Two bored people can choose different actions because one enjoys television, one plays guitar, one has a visitor and one has no usable instrument. A profession supplies schedules and competencies. A cultural profile can supply learned norms that a particular character follows to a stated degree. Demographic labels alone are too weak to determine an individual's next action.

For a middle-aged unemployed resident, model age, health, money, available time, household obligations, interests and local opportunities directly. Employment status removes a shift obligation; it does not imply laziness or a fixed personality. Ethnicity is not a shortcut to motivation. This produces more varied characters and makes the rules easier to explain and test.

5 A one-meter world with usable objects

Give the map integer meter coordinates. A cell describes basic terrain and movement information. Objects live in a separate registry: a 1 by 3 meter sofa is one object occupying three cells, not three unrelated sofas. Orientation, footprint and interaction points determine where a character may stand to sit, repair or move it.

Locations are named regions: a room, house, plot, park or forest. They may be nested or overlap. Alice can be inside a kitchen, inside a house and inside a residential district at once. Store region boundaries and references; do not copy the whole house into every occupied cell.

Layer	What it answers	Example
Meter cells	Can something stand or travel here?	Floor, wall, water, movement cost
Object registry	What thing occupies or affects this space?	Sofa, door, kettle, fallen tree
Region graph	Which places connect, contain or overlap?	Room to corridor to street
Action points	Where and how can an object be used?	One seat, two door sides, repair position
Dynamic reservations	Who may use it during this interval?	Alice has reserved the only toilet

Small cells do not require small updates everywhere

A sleeping district does not need every empty square revisited each second. Store terrain in chunks, such as 64 by 64 cells. Load detailed chunks where needed; represent distant roads as connected routes and empty land as regional data plus reproducible generation rules. Changes such as a damaged bridge remain persistent exceptions.

The initial prototype can use a flat map. Later, add a floor identifier and portals such as stairs before attempting a fully volumetric world. A one-meter grid alone cannot represent a narrow doorway, partial cover or bodies passing closely. Use interaction geometry or finer local navigation for those cases while keeping the meter grid as the broad spatial index.

Wear and damage use the same rule system. Mass affects carrying or moving; condition affects reliability; material affects damage; construction quality affects configured wear rates. These are separate properties. A door becoming broken may unlock a route, produce sound and change what nearby characters perceive in one accepted Beat.

6 Cheap routines still need social understanding

Procedural behavior can use internal state without running a language model. A small finite state machine can represent a work task. A utility policy can choose among leisure activities. A relationship rule can adjust trust after a witnessed broken promise. A short planner can connect a goal, such as eating, to prerequisites such as obtaining food and reaching a table.

Use the simplest mechanism that captures the intended consequence. Rules should explain their output using structured reasons: “Hungry; meal available; can afford it; chooses cafe.” A renderer can turn those records into readable process descriptions. It must not invent motives, dialogue or facts that the engine did not represent.

Observations are the bridge to belief

The world knows the door is broken. Alice learns this only if she sees it, hears it break or receives testimony. A rule can produce a perception record and then update an applicable belief. Rumor may be unreliable. A crowd's opinion is not automatically every member's opinion. The LWM's separation of world fact, perception, memory and belief remains in force.

Situation	Minimal procedural representation	When to ask for more detail
Colleague misses a promise	Witnessed failure; trust change; open obligation	Explanation is contested or relationship is pivotal
Dog hears a loud impact	Sound perception; arousal; approach or retreat tendency	Detailed encounter becomes relevant
Date proposes another meeting	Proposal, recipient response, availability, preference	Ambiguous intention or layered deception matters
Nightclub conversation	Noise, proximity, social availability, turn-taking	Exact dialogue and misunderstanding affect later choices
Lawyer prepares for a hearing	Task queue, deadline, skills, evidence access	Argument and knowledge claims become central

Theory of Mind can be procedural for bounded cases: Alice thinks Bob knows the door code because she watched him read it. Do not materialize every possible belief about every other person. Store only relevant attributions, their evidence and their depth. Missing detail remains marked as unmodeled. A later Storyteller may refine it only within the facts and perceptions already established.

7 When the Storyteller takes over

The procedural engine remains active across the world. The Attention Allocator promotes selected work into a Storyteller field when a player approaches, a critical NPC acts or a complex interaction requires richer reasoning. Promotion changes the method used to propose future actions; it does not create a new copy of the character.



Mechanical timers and world rules continue through the coordinator throughout.

Figure 2. Catch up, transfer authority, elaborate, then return current state. Mechanical services continue running through the same coordinator during the scene.

What must cross the boundary

The handoff packet includes current time, component versions, active actions, reservations, pending messages, effects and deadlines, relevant beliefs and memories, and the exact rule package version. The coordinator pauses the affected decision scope at a safe boundary and grants a new ownership epoch. A late proposal from the old owner is rejected.

The Storyteller may propose that Alice asks Devon for help. It cannot simply remove her fatigue, give her an item or make Devon know a secret. Movement, inventory and interaction rules validate the proposed action. In return, procedural rules must honor the scene's accepted promises, emotions and intentions after the field moves away.

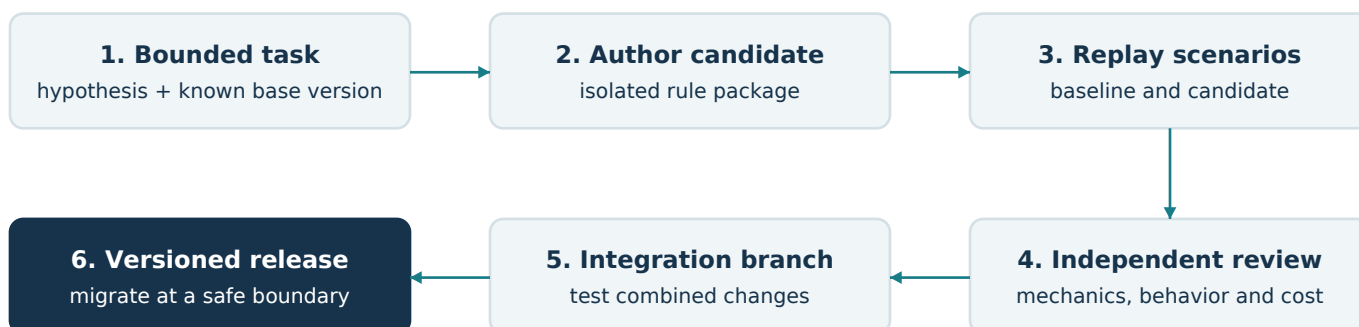
Same time, same person, same consequences

Different fields may advance in seconds or minutes. Before they interact, their relevant state must reach a common contact boundary. All earlier outward effects must already be known or explicitly ruled out. A background field cannot later invent a broadcast that people in an already completed scene should have heard. The LWM coordinator's boundary watermarks and causal ordering still apply [1].

If a Storyteller introduces a new trait or effect, it must map to supported state and mechanics. Otherwise the coordinator keeps that scope under detailed simulation or rejects the unsupported transition. Dropping an important new intention during demotion would break continuity even if the character's position stayed correct.

8 A workshop that improves the rules

The Rule Workshop operates outside the live world. A development agent receives a bounded task, such as “reduce implausible bathroom decisions before work” or “make territorial animal encounters less repetitive.” It inspects rule definitions, runs scenarios, studies traces and proposes a patch. Independent evaluators compare the candidate with a fixed baseline.



Failed candidates return for revision; accepted evidence remains reproducible.

Figure 3. Parallel experiments produce evidence, not automatic authority. Only reviewed, versioned packages reach a running world.

Parallelize experiments, not conflicting live edits

Different workers can improve occupations, household routines, navigation, animal behavior, objects or weather. Each works in an isolated branch or experiment snapshot tied to a known base version. A task allocator records ownership and dependencies so that ten agents do not unknowingly rewrite the same decision policy.

A proposed improvement includes the changed rules, the intended behavioral difference, scenario seeds, before/after traces, performance results and known trade-offs. Reviewers can test a combined candidate in a provisional integration branch. They do not need a person to manually inspect every numeric weight, but schema changes, privileged engine code and new write capabilities need stronger review than a bounded parameter adjustment.

The workshop is an iterative search process. More workers increase the number of experiments; they do not guarantee that realism improves. Shared blind spots, overfitting and interactions between individually good patches can make the combined simulation worse. The next page defines what counts as evidence of improvement.

9 What “better” should mean

Plausibility, entertainment, originality and emotional intelligence are different goals. A rule that makes everyone argue may increase drama while damaging everyday life. Keep mechanical validity as a hard gate, then evaluate behavioral quality along several dimensions. Let the chosen game setting determine acceptable trade-offs.

Dimension	Evidence to inspect	A failure example
Feasibility and continuity	Time, space, inventory, resource ownership	Two characters use the only toilet simultaneously
Motivational coherence	Need, goal, preference and obligation traces	Agent ignores an urgent need without a reason
Social and emotional fit	Perception, relationship, response and consequences	Comfort attempt ignores an explicit refusal
Variation and originality	Distribution of feasible actions across contexts	Everyone always watches television
Narrative usefulness	Opportunities and consequences over an episode	Surprise events erase earlier player choices
Cost and stability	CPU, memory, path requests, queues, long-run health	Better dialogue causes a tenfold scheduling cost

Strong language models can review structured episodes and readable transcripts. Their role is to assess coherence and authored behavior, not certify scientific truth about people. Research on LLM judges identifies position, verbosity and self-preference biases [6]. Compare equally formatted traces, hide candidate identity, swap presentation order and use independent judges plus human calibration samples.

A concrete evaluation question

Does the new bathroom rule reduce inexplicable lateness while preserving urgent need relief, reasonable queue behavior and commitments? Run paired baseline/candidate scenarios with the same initial worlds and controlled random streams. Include an occupied toilet, a long commute, a sick child, a mobility constraint and a job with flexible hours. Also run unrelated scenarios to detect side effects.

The evaluator must see the state before the action, feasible alternatives, decision evidence and downstream results. Beautiful prose cannot compensate for a missing perception or impossible object transfer. Numerical distributions and invariant checks remain necessary even when every judge prefers the candidate's story.

10 Scale depends on what “simulate” means

A million stored people, a million people updated once a simulated minute, and a million people navigating every 0.2 seconds are radically different workloads. The map's area is another independent quantity. A vast empty landscape can be cheap when generated on demand; a crowded station can be expensive within one square kilometer.

The following are **planning configurations for an optimized CPU engine**, not benchmarks. Most actors use coarse scheduled updates. Only the detailed subset gets frequent local decisions. Python objects with large dictionaries will not achieve these memory figures. Appendix C gives every storage assumption and the processing sensitivity model.

Installed RAM class	Example detailed terrain	Resident people / animals	Detailed procedural actors
8 GB	25 km2	100,000 / 20,000	200
16 GB	100 km2	400,000 / 100,000	800
32 GB	250 km2	1,000,000 / 250,000	2,000

With the stated compact records, objects, building counts and headroom, these examples use about 1.94, 7.08 and 17.32 GiB of simulation memory. They leave separate system and graphics allowances. The actor counts are conditional targets for a compiled, measured implementation; they are not promises for the first Python prototype or for every CPU in that RAM class.

What changes at continent scale

For a chosen continent area of 10 million km2, allocating eight bytes for every meter cell requires **80 TB before actors or navigation**. Streaming avoids this dense allocation. With 10 million people, 10 million animals, 100 million objects and only 1,000 km2 of detailed terrain resident, the worked current-state budget is about **218 GiB including headroom and a system allowance**. A 256 GiB server is a tight starting configuration; 512 GiB leaves room for larger state and bursts. History and replicas are additional.

Compute can be modest for coarse schedules and enormous for continuous detailed action. A whole continent at full scene detail is not justified by the cheap background estimates. Storyteller inference and the development workshop also require their own compute, potentially GPUs or remote model services, even when the world runtime is CPU-only.

11 A prototype that can grow

Begin with a **256 by 256 meter neighborhood**, 32 people, eight animals, a few houses, one workplace and roughly 300 objects. Display it with simple top-down pixel art. The purpose of the display is to reveal behavior: blocked routes, queues, strange pauses and social interaction. The same simulation must run without graphics for fast tests.

Stage	Add	Demonstrate before expanding
First block	Needs, movement, furniture, schedules, doors	A stable day with no impossible overlap or lost actions
Small neighborhood	Households, obligations, work, animals	Plausible choices and inspectable causes across several days
Rule Workshop	Candidate packages and independent evaluation	One improvement survives unrelated regression scenarios
Storyteller bridge	Shared Beats and ownership handoff	A new scene preserves prior procedural history
District and city	Chunk streaming, route graph, compact arrays	Cost grows with active work rather than total map area

Python can coordinate the prototype, load rule data, run evaluations and render the map. Numerical arrays hold frequent state. Tight loops can later move into compiled kernels, such as Numba-supported numeric functions, or a native extension without changing the external rule contracts [2, 3]. Do not build the first version around millions of nested Python instances and expect a compiler switch to solve every bottleneck.

What the player should notice

Leave Alice during breakfast, visit Devon, then return. Alice should have acted within her schedule and knowledge. If a rule made her miss work, the trace should explain why. Interrupt an off-screen conversation with a call; the resulting state should be the same state the player later encounters. Damage a door; routes, sound and perceptions should react consistently.

The prototype earns the right to grow by explaining its consequences. The scale-up path is to preserve those contracts while reducing unnecessary work, not to replace characters with unrelated approximations each time the camera moves.

12 The system in a twenty-minute conversation

The design has three cooperating parts. The **world runtime** cheaply advances people, animals, places and objects. The **Storyteller layer** spends richer reasoning on selected interactions. The **Rule Workshop** improves the reusable rules through controlled experiments. One coordinator accepts all world changes, and one history makes them inspectable.

Follow one complete chain

Devon feels bored. His preferences make guitar practice attractive, but a household rule says the room is quiet while a baby sleeps. He can use another room, choose a quiet activity or postpone practice. A rule proposes a feasible plan and reserves what it needs. Time advances; needs change; the plan can be interrupted. The accepted Beat records the action and its consequences.

Later a coding agent notices that too many people in test neighborhoods practice guitar through the night. It changes the noise-context rule, not Devon's established past. An independent evaluator tests the patch across households, night shifts, musicians and people without another room. The combined rule package is staged, measured and released. New decisions use the new version after a safe migration boundary.

Five commitments worth retaining

Commitment	Practical meaning
One world history	Procedural and Storyteller updates cannot invent competing facts
Explicit rules and sources	A decision can identify the rule, inputs and evidence behind it
Components over endless subclasses	Roles, needs and preferences can change independently
Selective computation	Cheap routines, scheduled events and detailed scenes coexist
Evidence before integration	More agent-written code is not itself an improvement metric

Technical reading map

Appendix A defines data, rules, methods and spatial representation. Appendix B covers runtime scheduling, authority and computational complexity. Appendix C contains the capacity estimates for laptops, gaming PCs and a continent. Appendix D specifies the parallel workshop, evaluation gates, merge process and rollout. Appendix E gives the Python prototype structure and acceptance scenarios.

All numerical rules below are explicit design examples. They can be replaced while preserving the interfaces. The goal is an implementation brief that a coding agent can act on and an interested reader can challenge, rather than a collection of unexplained AI behaviors.

A1 Classes for authors, components for the engine

A shallow authoring hierarchy gives each entity sensible defaults. Composition handles independent variation. The runtime compiles these definitions into component schemas and system subscriptions. An entity keeps its stable ID when its component set changes.

```
Entity
  Actor -> Human, Animal, SyntheticActor
  Object -> Furniture, Tool, Door, Consumable
  Location -> Room, Building, Plot, Region
  Process -> WeatherCell, DecayProcess, HazardEvent

Human instance Alice
  body_profile: human_default
  life_stage: adult; age_world_years: 42
  components: Needs, Skills, Preferences, Perception,
              Beliefs, Goals, Schedule, Inventory, Relations
  roles: caregiver, employee
  individual_overrides: authored preference parameters
```

Inheritance and override resolution

Resolve base type, species/body profile, life-stage profile, context roles, then individual overrides. This order applies only to fields that declare an override operation. Other fields may add, multiply, take a maximum or use a dedicated resolver. Two packages assigning incompatible absolute values to the same property produce a build error. Do not use import order as a conflict policy.

Species profiles supply capabilities and physiology assumptions. Life stages govern development and available actions. Roles attach tasks and permissions. Culture profiles describe learned norms and institutions with explicit individual adherence. Temporary effects attach duration and combination rules. A single subclass for every combination would multiply the number of classes and make employment or aging a disruptive identity change.

Runtime storage

Use stable integer handles into component tables, with generation counters to reject references to deleted entities. Frequently processed fields become contiguous numeric columns: position, needs, current action and next update time. Rich but infrequent state sits in separate records. Unity's ECS documentation illustrates the general idea of grouping entities by their component sets and storing component arrays together [4]; this proposal does not require Unity or its exact chunk format.

A2 Minimum shared data contracts

Contracts describe what must be represented, not a required database vendor. Every mutable record has a version, and every accepted change references a Beat. Numeric fields specify units, bounds, defaults and missing-value behavior.

Record	Required content
EntityRecord	ID, kind, schema version, component handles, lifecycle, owner epoch
ActorState	Position or route progress, supported needs, goals, action state, capabilities, accessible evidence references
ObjectState	Prototype ID, transform, footprint, mass, condition, ownership, container, affordance and reservation references
LocationState	Geometry reference, hierarchy, entrances, access policy, capacity, opening calendar, occupancy view
NeedDefinition	Unit, urgency direction, rate policy, thresholds, action relief rules, supported body profiles
Goal / Commitment	Holder, target, prerequisites, priority policy, deadline, progress, status, related entities
ActionInstance	Definition version, actor and targets, start/end bounds, phase, reservations, progress, interruption policy
RuleDefinition	Typed condition, read/write declarations, outputs, timing, randomness policy, resource bounds, package version
EffectInstance	Source, target, onset, expiry, modifiers, stacking rule, causal evidence
ScheduledEvent	Due logical time, event type, payload, owner, deduplication ID, causal parent
RulePackage	Immutable content hash, compatible schemas, dependency versions, tests, migration plan, release manifest

Time, space and quantities

Store world time in integer units with a documented precision, for example milliseconds, plus causal microsteps for same-time ordering. Use meters for distance, seconds for duration and explicit units for mass and rates. A normalized need value is unitless but its growth rate is per world second. Locale affects display, not stored units.

An unknown belief is not a false belief. An unmodeled emotion is not an unchanged emotion. Coverage flags distinguish those cases. A compact background actor stores bounded current beliefs and open obligations; longer experience stays in referenced history. The rule contract declares when missing detail requires promotion instead of a fabricated default.

A3 A rule format that agents can safely extend

Prefer a declarative rule language for most changes: typed comparisons, Boolean conditions, bounded expressions, known queries and approved action/effect names. It can be stored as structured documents and compiled before release. Arbitrary code execution is unnecessary for editing a need threshold or adding a profession schedule.

```
RuleDefinition: human.prepare_for_work / version 3
  applies_to: Actor with Employment, Needs, Mobility, Schedule
  trigger: shift_approaching OR urgent_need_changed
  reads: own needs, known shift, known routes, known affordances
  proposes: UseToilet OR DepartForWork OR SeekAlternative
  decision_scope: actor.activity_choice
  duration_policy: supplied by the chosen ActionDefinition
  priority_band: ordinary_commitment; emergencies may interrupt
  randomness: one named decision draw; no retry resampling
  budget: candidate_limit=24; query_radius=local; no world scan
  writes: proposed action only; resolver applies world effects
  explanation: input references + scores + chosen alternative
  dependencies: needs.v2, travel.v3, reservations.v1
  tests: private_toilet, occupied_toilet, late_shift, no_route
```

Rule families

State-rate rules advance gradual quantities. Eligibility rules decide what can be attempted. Utility rules rank feasible actions. Selection rules introduce bounded variation. Transition rules apply success, failure or interruption effects. Observation rules determine what an actor can perceive. Belief rules interpret accessible evidence. Schedule rules manage deadlines. Process rules advance objects and environmental systems. Invariant rules reject impossible results.

Each family has declared authority. A utility rule cannot directly move an object; a narrator cannot invent a new damage value; a perception rule cannot copy private belief state from another actor. New functionality that exceeds the declarative language uses a reviewed engine plugin with the same typed inputs, effects and cost limits.

A rule package is rejected before simulation if it references an unknown component, mixes units, contains an unbounded query or lacks required migration and test data. Declarative rules reduce the review surface; they do not remove the need to test their interactions.

A4 Actions, methods and multi-actor interaction

An action is a protocol that lasts through time. Its definition supplies preconditions and mechanics; an ActionInstance records one attempt. Separate the actor's intention from what the world allows to happen.

Method	Inputs and result	Required behavior
available_actions	Actor perspective, local affordances -> candidates	Return bounded, known possibilities
evaluate_preconditions	Candidate, versions, access -> result	Explain failure or missing information
score	Feasible action, needs, goals -> score terms	Preserve interpretable contributions
reserve	Resources, interval, owner -> reservation	Resolve competition atomically
start	Accepted intent, preconditions -> action state	Charge start costs once
advance	Action state, elapsed time, events -> proposed delta	Stop at interruption or outcome boundaries
interrupt	Reason, current progress -> cleanup and effects	Release reservations; preserve spent resources
complete	Final checks, random draw -> outcome	Apply one result and one causal trace

A conversation is a joint protocol

Approach, request attention, accept or decline, exchange turns, end. Each participant owns a decision; neither can force the other's response. Shared resources include attention, space and possibly an exclusive device. A date, negotiation or service encounter can specialize this protocol with different goals and norms without copying its lifecycle mechanics.

Reservations do not guarantee eternal success. Before an effect occurs, recheck that the target still exists, is reachable and remains usable. A toilet can break after it was reserved. A door can become blocked during travel. Failed actions produce explicit outcomes and new planning opportunities rather than spinning in immediate retries.

Maintain a cooldown or commitment margin so small score changes do not make an actor alternate between television and guitar every update. Use declared emergency interrupts for urgent needs and hazards. Costs already incurred, such as travel time or consumed material, remain in history if an action is canceled.

A5 Needs, probability and time-step independence

The simple need equation on page 5 can be integrated lazily. Store a value, rate and last-update time; calculate the present value when it is read or a threshold is due. Schedule the next threshold crossing. When eating changes digestion or a medicine-like fictional effect changes attention, store a new rule segment with its own effective time.

Probability belongs to an opportunity

An 80% toilet-first rule describes a conditional decision among feasible alternatives. Sample it once when a decision opportunity opens. Keep the result across retries and harmless time-step subdivisions. Otherwise making Beats shorter would create more opportunities to obtain the same outcome.

For a random event with a constant hazard rate λ per second, the chance within duration dt is:

$$P(\text{event during } dt) = 1 - \exp(-\lambda * dt)$$

This is appropriate only for that chosen hazard model. If conditions change, integrate the changing rate or split the interval. A per-tick 1% probability cannot be reused at arbitrary Beat lengths. For action choice, a normalized softmax over feasible scores is one option; specify its temperature, numerical stability and fallback when there are no candidates.

Reproducible randomness

Name random streams by world seed, entity or interaction ID, decision ID, rule version and draw purpose. Worker completion order must not decide which random number Alice receives. Parallel NumPy streams can be derived using SeedSequence, but stable stream-to-task mapping is still the application's responsibility [5]. A retry of the same proposed action reuses its draw; a genuinely new opportunity receives a new ID.

For paired experiments, align streams for unchanged decisions and report where candidate behavior diverges. Common random numbers improve comparison where paths remain comparable; they do not make divergent stories identical. Replaying accepted Beats means replaying their stored outcomes, not regenerating random decisions from prose.

A6 Objects and a worked percentile check

Objects expose affordances and response functions. A door has a footprint, hinges or slide direction, open/closed/locked state, material, integrity and relevant permissions. A hammer has mass, grip requirements, condition and compatible action effects. Ownership and physical custody remain separate.

A deliberately simple game mechanic

For a fictional door-damage test, let the effective strike skill be 35 on a 0-100 scale. Draw an integer uniformly from 1 to 100 and accept a successful effective strike when roll ≤ 35 . This yields exactly 35%; using roll < 35 would yield 34% with this die convention. This is a generic percentile mechanic, not an implementation of a named role-playing ruleset.

Step	Example value	Result
Preconditions	Hammer held; door in reach; two seconds available	Attempt is feasible
Effective strike	Roll 27 against threshold 35	Success
Raw damage	Uniform integer 8-14; draw is 11	11 damage units
Material resistance	3 damage units	Net damage = $\max(0, 11 - 3) = 8$
Integrity	7 before the strike	$\max(0, 7 - 8) = 0$; broken
Other consequences	Duration, effort, sound, tool wear	Scheduled or logged by their configured rules

A stationary door need not require a miss chance in every game. A designer may use certain contact with uncertain effective damage. The important point is that the selected mechanic has explicit units, duration, probability and state changes.

When the door breaks, invalidate affected navigation links and re-evaluate waiting actors. Nearby observers receive sound or sight perceptions according to access. A distant character does not learn the result automatically. Repair has its own prerequisites, consumed materials, progress and resulting condition; it must not restore more material or value than the authored rules permit.

Routine wear should depend on elapsed time, usage and exposure. Do not decrement integrity simply because a field happened to receive more updates. A barely used chair in a detailed player field should not age faster than the same chair off screen.

A7 Geometry, buildings and action affordances

Use a global coordinate system plus chunk-local cell indices. At one-meter resolution, a 64 by 64 chunk has 4,096 cells. A cell can contain compact terrain flags and references into shared structures. Multiple objects and actors may be associated with a cell through a spatial index rather than one exclusive object-ID slot.

Footprints and access

Store each object's occupied cells or polygon, height class, orientation and interaction anchors. A sofa may block three cells while offering several seating anchors. Doors block passages conditionally. Access anchors must remain reachable, with a clearance rule appropriate to the actor's body. This prevents a valid-looking furnishing layout from silently trapping someone.

Locations use polygon or cell-mask geometry with holes, floor IDs, containment and overlap relations. A park can contain a pond; a building can contain several households. Permissions, ownership, occupancy and a region's geometric membership are separate properties. Cache derived membership but invalidate it when geometry changes.

Navigation at more than one scale

Inside a loaded room, use local walkable geometry. Across buildings, use entrances and road links. Across a district, use a route graph with travel time and capacity. Store route progress and estimated arrival; refine the local path before observable contact. Every coarse route must respect obstacles and closures that could change arrival or interaction.

At continent scale, terrain generation describes unchanged empty areas and sparse edits describe exceptions. A bridge destroyed months ago must remain destroyed when its chunk reloads. Store the generator version and seed with edits; upgrading the generator requires an explicit migration so that landmarks do not move underneath recorded history.

Meter resolution describes addressability, not permanent residency. To promise persistent state for every one-meter cell, even when each cell may change independently, the system must eventually store those changes. Sparse storage saves memory only while unchanged or compressible areas remain common.

A8 Animals, weather and other processes

Animal profiles combine senses, movement, needs, species-appropriate action opportunities and individual learning. A dog, bird and grazing animal should not share an identical work-and-leisure policy with different sprites. Pack membership, territory, care relationships and threat response can be components shared where appropriate.

Process	State and method	Appropriate update strategy
Animal routine	Needs, location, shelter, social relations	Threshold events plus local decisions
Day and night	World calendar and region lighting function	Calculate directly from time
Weather	Coarse cells, forecast/state, fronts, exposure	Regional updates; local effects on demand
Object aging	Exposure, use, material, damage history	Integrate rates; schedule failure opportunities
Fire or flood	Region, intensity, propagation constraints	Refine where spread or contact is imminent
Plant or crop growth	Stage, resources, environment, time	Slow progression with exceptional-event interrupts
Group or company	Members, accounts, contracts, schedules	Aggregate accounting plus consequential events

Weather is not an actor merely because it changes. Model it as a process that affects exposure, travel, resources and perception. A thunderstorm can be regional while a falling branch becomes a local object event. Fine physical fluid or atmospheric simulation is a different, much larger workload than these game abstractions.

Groups have shared records and official decisions, not automatically one shared mind. A company rule may publish a shift schedule; an employee still needs a valid information path to learn a change. Distribute public notifications through normal channels. A faction director's authorial plan remains separate from the faction's in-world knowledge.

Coupled shocks are important scale tests. An evacuation or sudden storm can wake thousands of actors and invalidate many routes together. Admission control, priority queues and regional approximation must cope with this burst. A calm-day throughput estimate says little about the worst crowded emergency.

B1 The CPU runtime and its interfaces

Separate the authoritative state store, scheduler, spatial queries, decision systems, action resolver and presentation. The compiler translates approved rule packages into bounded operations over components. A readable authoring interface does not require per-entity Python method dispatch inside every hot loop.

```
RuleEngine.propose(snapshot, due_events, rule_manifest)
  -> BeatProposal + next_wakeup_hints + diagnostic_trace

Coordinator.validate_and_commit(proposal, ownership_epoch)
  -> CommitReceipt OR structured_rejection

SpatialService.query(bounds, filters, snapshot_version)
  -> bounded entity/affordance references + coverage

NarrationService.describe(accepted_beat_refs, viewpoint)
  -> text linked to recorded facts; no new world effects

EvaluationService.compare(baseline, candidate, scenario_manifest)
  -> invariant results + metrics + judged episodes + costs
```

One useful execution cycle

Drain due messages and timers in causal order. Update only the required rate-driven components. Collect bounded action candidates for actors whose decisions are due. Resolve reservations and conflicts. Advance actions to the next safe boundary. Derive perceptions and eligible internal-state changes. Validate and commit the Beat and outgoing messages together. Schedule the next necessary wakeups.

Entity-centered history can be indexed asynchronously, but required current facts use a consistent version. A stale search result cannot be allowed to restore an old owner or belief. Deterministic replay stores rule hashes, outcomes and arbitration decisions; it does not depend on a language model reproducing the same description.

For the first implementation, use one coordinator and whole-entity write ownership. Later, separate safe component scopes and shard by region. Sharing immutable rule code is easy; coordinating shared mutable state is the difficult part. Profile serialization, queues and memory traffic as well as decision arithmetic.

B2 Scheduling: update what is due

A full scan of all actors on every render frame wastes work. A scheduler stores the next time an actor, object or process requires attention. Sleeping actors may wake at a need threshold or alarm. Travel can wake at arrival or a route change. An effect wakes at expiry. Rendering interpolates compatible motion independently.

Runtime mode	Example cadence	State guarantee
Detailed local procedure	0.2-2 seconds, plus interrupts	Fine actions and interaction constraints
Routine individual	30-300 seconds or deadlines	Named identity, needs, tasks and outward effects
Dormant individual	Analytical intervals plus mandatory events	Exact commitments; only constrained private detail remains open
Regional aggregate	Minutes to days	Population/resource totals and exported consequences
Storyteller scene	Budgeted Beats plus mechanical services	Relevant speech, perception and internal state

These are policies, not an automatic mapping to L0-L5. A remote contract negotiation may be slow in time but semantically detailed. Aggregating unnamed wildlife is different from preserving ten million named animal histories; capacity claims must state which is used.

Queue design

A heap provides predictable logarithmic insertion and removal. Timing wheels can make common bounded timer operations cheaper but need explicit resolution, overflow and cancellation handling. Coalesce redundant wakeups by entity and reason. Lazy canceled entries require cleanup so the queue cannot grow forever.

Respect external effects

Analytically skipping sleep is valid only if no relevant interruption was skipped. An alarm, visitor, hazard or incoming message cuts the interval. Coarse policies must enumerate or rule out all outward effects up to the receiver's safe frontier. Zero-delay dependency cycles may require joint advancement. One fast background pass cannot certify an arbitrary open world in which any actor may invent an unregistered effect on anyone else.

On overload, reduce optional detail, limit new speculative work or slow world time. Do not drop critical timers and still call the interval simulated. Report the coverage that actually advanced.

B3 Complexity and the costs that dominate

Let N be resident actors, D the actors with due decisions, K the bounded candidates per decision, P path requests, V explored navigation nodes and E navigation edges. Let Q be queued timers, C resident cells and X committed changes. The expressions below describe work, not measured seconds.

Operation	Approximate cost	Scaling risk
Full actor scan	$O(N)$ per scan	Paying for inactive actors repeatedly
Due decision evaluation	$O(DK)$ plus queries	Unbounded candidate lists or expensive predicates
Heap scheduling	$O(\log Q)$ per queue operation	Millions of redundant or stale timers
Spatial broad-phase query	Expected local buckets + returned candidates	Dense crowds collapse locality benefits
All-pairs social comparison	$O(N \text{ squared})$	One million actors imply about 500 billion pairs
Heap-based graph search	Up to $O((V + E) \log V)$ per search	Too many long or invalidated paths
Numeric component pass	$O(\text{number of touched values})$	Memory bandwidth and copies dominate arithmetic
Beat logging	$O(X)$ data plus indexing	Retaining routine deltas at tiny intervals
Dense map allocation	$O(C)$ memory	Area grows quadratically with map side length

Make expensive work explicit

Cache route segments, share suitable paths, use region-level travel, and replan when necessary. A 2 ms path request at 1,000 requests per second consumes two CPU-seconds per second before any other work. A 100 ms crowded search can cause visible stalls even if the average looks excellent.

Query social candidates through spatial locality, relationships and active events. Store a bounded number of meaningful relations by default; do not allocate a full person-by-person matrix. A global broadcast can be one shared event with scoped recipient processing, but consequences still have a cost when recipients react.

Parallel speedup is limited by serial coordination, memory bandwidth, cross-region messages and imbalance. Hardware threads are not interchangeable with full physical cores. Report p50, p95 and p99 update latency, queue delay and total CPU time, not just average actor updates per second.

B4 Storyteller handoff as a transaction

The promotion contract is an authority transfer, not a prompt containing a rough biography. It specifies the exact state and time that the next engine must continue. Storyteller fields may span several procedural regions, so the coordinator acquires all relevant scopes before enabling writes.

```
HandoffPacket
  entities_and_components, common_logical_time
  snapshot_version, old_epoch, new_epoch
  active_actions, progress, reservations, pending_inputs
  effect_deadlines, open_commitments, required_world_rules
  agent_specific_perceptions_memories_beliefs
  outward_effects_sealed_through, unresolved_detail_constraints
  pinned_rule_manifest, semantic_profile, component_overrides
  intended_return_contract
```

Promotion protocol

Identify the interaction set and dependencies. Stop affected writers at a common safe boundary. Process required inputs and advance coarse state to contact. Cancel uncommitted proposals beyond that point. Transfer ownership with a new epoch and provide perspective-specific packets. Mechanical services continue only within their declared authority, submitting changes through the same coordinator.

Demotion protocol

Finish or serialize active actions, promises, reservations and effects. Check that all future-relevant internal state can be represented procedurally. Map free-form intentions to supported goal types with explicit targets and deadlines; retain the authored text as evidence. If mapping is ambiguous, keep the relevant field active until resolved. Return component ownership and future queues, not a narrative summary alone.

Three fields that interact through an A-B-C chain require one coordinated group even when A and C do not touch. Split only when shared decisions end and message routing is transferred. A story about a ring must never create two authoritative custodians. A coarse-to-fine refinement adds permitted detail while retaining accepted transitions and external effects; it does not apply the old transfer twice.

C1 How the estimates should be read

Capacity has at least four dimensions: map area, resident state, due work per second and retained history. The tables below separate them. **Every processing cost is an assumed sensitivity input, not a measured result.** Exact arithmetic can show whether a design is plausible; only a representative benchmark can establish capacity for a particular machine.

Units and assumptions

GB and TB mean decimal bytes. KiB and GiB use powers of 1,024. For capacity arithmetic, “8/16/32 GB RAM class” is treated as 8/16/32 GiB addressable RAM; real operating-system reporting and integrated graphics reservations vary. Simulation budgets are 3, 8 and 20 GiB respectively, after a conservative system/graphics allowance. Do not also load local LLM weights into that allowance.

Assumption	Planning value	Included / excluded
Terrain cell	8 bytes	Compact terrain/navigation fields; no full object or search workspace
Routine human	8 KiB	Bounded current state, relations, beliefs, queue references
Routine animal	2 KiB	Simpler bounded state; species can require more
Object	128 bytes average	Compact instance; shared prototypes; large inventories extra
Building	2 KiB average	Registry and region metadata; detailed interiors extra
Detailed actor supplement	64 KiB	Temporary planner and fine state; no LLM prompt cache
Working headroom	50% of subtotal	Indices, allocator overhead and bounded transient work

These are design budgets, not measured sizes of Python instances. Histories, backups, textures, replicas and unbounded social graphs are excluded. NumPy's nbytes describes array element storage rather than all associated object overhead [2]; measure whole-process resident memory as well as array sizes.

For processing, test a warm actor update cost *c* of 1, 5 and 50 microseconds. A separate detailed update assumes 100 microseconds before pathfinding, persistence and language-model work. These values bracket proposed kernels; they do not follow from CPU frequency or vendor peak benchmarks.

C2 The price of a one-meter map

A square kilometer contains one million one-meter cells. With A in square kilometers and b bytes per cell, dense map memory is A times 1,000,000 times b. This is only the base grid. Objects, actors, paths, graphics and history consume additional memory.

Dense area	Side length if square	Grid at 8 B/cell	Grid at 32 B/cell
0.0625 km2	250 m	0.48 MiB	1.91 MiB
1 km2	1 km	7.45 MiB	29.80 MiB
25 km2	5 km	0.186 GiB	0.745 GiB
100 km2	10 km	0.745 GiB	2.980 GiB
1,000 km2	31.6 km	7.451 GiB	29.802 GiB
10 million km2	3,162 km	80 TB / 72.76 TiB	320 TB / 291.04 TiB

Why a sparse continent is different

A world can have meter coordinates everywhere without allocating a cell record everywhere. Procedural generation plus coarse regional maps supplies unchanged terrain. Persist edited chunks, named locations and route graphs. A 64 by 64 chunk at eight bytes per cell holds 32 KiB of base cells. One thousand km2 of resident detailed cells remains about 7.45 GiB regardless of how far apart those chunks are.

Cold changes still occupy disk. If eventually every meter is edited independently, sparsity disappears. A dense 3D volume scales with height as well as area, so even modest vertical depth can multiply the terrain cost. Floor layers and portals are a better early fit for buildings than allocating all air as voxels.

Buildings are not interchangeable with households

The continental example uses 100,000 building records to match “many tens of thousands” in the design brief. Ten million people would average 100 people per building before allowing for empty or nonresidential buildings. That is a dense settlement assumption, not a normal detached-house city. A million smaller buildings would add about 1.72 GiB of raw registry data under this budget, but their interiors, furniture, roads and navigation could add much more.

C3 Worked local memory configurations

The following configurations satisfy the declared simulation-memory budgets. Counts are simultaneous, so readers do not have to combine unrelated maxima. All terrain listed is resident at one-meter resolution. Human and animal counts include the detailed subset; its 64 KiB supplement is additional.

Quantity	8 GB class	16 GB class	32 GB class
Simulation budget	3 GiB	8 GiB	20 GiB
Detailed terrain	25 km2	100 km2	250 km2
Humans	100,000	400,000	1,000,000
Animals	20,000	100,000	250,000
Objects	1 million	4 million	10 million
Buildings	5,000	20,000	50,000
Detailed actors, all drawn from humans	200	800	2,000
Raw current-state subtotal	1.128 GiB	4.551 GiB	11.378 GiB
After 50% headroom + 0.25 GiB fixed	1.942 GiB	7.077 GiB	17.318 GiB

```
M_raw = 8 * resident_cells + 8192 * humans + 2048 * animals
        + 128 * objects + 2048 * buildings + 65536 * detailed_actors
M_sim = 1.5 * M_raw + 0.25 GiB
```

Interpretation

These are memory-feasible optimized designs, not tested whole-system limits. The smallest build should still begin with dozens of actors. A pure Python reference engine can be much larger per actor because every dictionary, string and list adds overhead. If the human record grows from 8 KiB to 32 KiB, the one-million-person row gains roughly 22.9 GiB before headroom and no longer fits.

Actor memory also grows when relationships, goals, inventories and beliefs are left unbounded. A rare richly represented character may require far more than the average. Enforce budgets and retain older detail in indexed history, while promoting current state when the world requires it. The system must report when it has exceeded its assumed schema rather than hiding the extra memory in a generic “AI” category.

C4 CPU work on a laptop or gaming PC

Let U be actor updates per real second, c the mean CPU time per update, p the usable physical-core equivalent and η the fraction of those cores assigned to this work. The necessary budget condition is $U \text{ times } c \leq p \text{ times } \eta$. It is not sufficient: pathfinding, logging, memory traffic and latency can still dominate.

```
updates_per_real_second = actors * world_time_speed / world_interval
CPU_seconds_per_second = updates_per_real_second * cost_per_update
```

Workload at normal world speed	Update rate	Assumed cost	Compute-only cores at 50% budget
10,000 actors every second	10,000/s	5 us	0.10
1 million actors every minute	16,667/s	5 us	0.17
10,000 detailed actors at 5 Hz	50,000/s	100 us	10
1 million detailed actors at 5 Hz	5 million/s	100 us	1,000

Applying this to the three local configurations

Use 60-world-second routine human updates, 120-second animal updates and 5 Hz detailed updates. After removing detailed humans from the routine count, the configurations require approximately 1,830 / 7,487 / 18,717 routine updates per second and 1,000 / 4,000 / 10,000 detailed updates per second. At the assumed costs, their actor-kernel CPU demands are 0.109 / 0.437 / 1.094 CPU-seconds per second. At 10 times world speed, multiply each by ten before other costs.

Thus a million mostly scheduled residents may be feasible on a capable optimized desktop, while a million continuously detailed residents is a different scale entirely. As hardware anchors, Intel's Core Ultra 7 258V specifies eight mixed-type cores, while AMD's Ryzen 9 9950X specifies 16 cores and 32 threads [7, 8]. These specifications do not establish simulation throughput, and low-power cores must not be counted as equal-speed performance cores.

For planning, reserve only 2-4 strong-core equivalents on a laptop and 4-8 on a gaming desktop until measured. Start an interpreted Python build with dozens to hundreds of detailed actors; treat thousands as a profiling target. Battery mode, cooling and rendering reduce sustained capacity. CPU-only here excludes model inference and the rule workshop.

C5 A continent with twenty million actors

Choose a 10-million-km2 world with 10 million humans, 10 million animals, 100 million objects and 100,000 building records. Most terrain is generated or stored cold; 1,000 km2 is resident in detail. Twenty thousand humans have the detailed-state supplement. These are scenario parameters, not geographical facts.

Current-state component	Calculation	Memory
Humans	10 million x 8 KiB	76.294 GiB
Animals	10 million x 2 KiB	19.073 GiB
Objects	100 million x 128 B	11.921 GiB
Buildings	100,000 x 2 KiB	0.191 GiB
Resident meter grid	1,000 km2 x 8 B/m2	7.451 GiB
Detailed supplements	20,000 x 64 KiB	1.221 GiB
Regional indices, routing and queues, base allowance	Fixed planning allowance	8.000 GiB
Subtotal	Sum of the above	124.150 GiB
With 50% working headroom	1.5 x subtotal	186.226 GiB
With 32 GiB system/service allowance	Current-state planning total	218.226 GiB

What kind of deployment follows?

For the coarse workload, a large-memory server with 256-512 GiB and dozens of CPU cores is a plausible benchmark starting point. It is not a demonstrated minimum. A 512 GiB configuration provides safer room for dense places, larger records and experiments; replicas and additional whole-world snapshots are extra. A distributed deployment can divide this state across regional workers while preserving global IDs and boundary messages.

Modern server classes can supply many cores and memory channels; the EPYC 9755, for example, specifies 128 cores and 12 DDR5 channels [9]. This is a hardware reference, not a purchasing recommendation or a claimed match to the model. A cluster may be needed for faster-than-real-time runs, resilience, heavy navigation or many detailed regions even when a single server can hold the compact state.

Full continental interiors, unlimited personal histories and all cells resident would invalidate this budget. The dense terrain alone would be 80 TB. The next page shows that update cadence and history retention may matter more than the actor record size.

C6 Continental compute, communication and history

With humans reconsidering routine state every 60 world seconds and animals every 120, twenty million actors require about 250,000 routine updates per world second. The first rows below isolate this coarse kernel. A detailed subset must be removed from that routine count and budgeted separately.

Continental routine mode	Updates / real second	Cost 1 / 5 / 50 us	Core equivalents at 50% budget
Normal world speed	250,000	0.25 / 1.25 / 12.5 CPU-s/s	0.5 / 2.5 / 25
60x world speed	15 million	15 / 75 / 750 CPU-s/s	30 / 150 / 1,500
All 20 million every second, normal speed	20 million	20 / 100 / 1,000 CPU-s/s	40 / 200 / 2,000

Twenty thousand detailed actors at 5 Hz and 100 us add about ten CPU-seconds per real second at normal speed, before navigation. Ten thousand route requests per second at 2 ms each add another twenty. Dense meetings, disasters and cache misses can raise these costs sharply. Cross-region traffic is e boundary events per second times b bytes per event; 100,000 events/s at 256 bytes is 25.6 MB/s payload before replication, protocol overhead and latency.

The history budget is separate

If all 20 million actors emit one retained 192-byte event every world minute, that is roughly 333,333 events/s and **5.53 TB per simulated day**, before indexes or replication. One event per actor per world hour is still **92.16 GB per simulated day**. At accelerated world speed, the real-time write rate rises proportionally.

Store meaningful changes and analytical intervals instead of repeated unchanged values. Keep evidence required for obligations, perceptions and future causality. Summaries help retrieval but do not automatically replace authoritative history. If a product promises exact replay forever, storage grows accordingly. If old records are compacted or expired, define which historical queries remain exact and retain an audit trail of that policy.

Compression and cold storage can help, but no compression ratio is assumed here. Disasters and highly active days may produce much more history than ordinary schedules. Current-state RAM, history capacity, backup bandwidth and replica memory must appear as separate budget lines.

D1 Scaling the Rule Workshop to many agents

A development swarm is a pool of stateless workers around a persistent experiment service. Agents retrieve bounded tasks, produce candidate packages and leave evidence. They do not all clone a running continent into memory or directly write to the main branch. Simulations can use small targeted worlds, then larger representative regions.

Service	Responsibility	Scale boundary
Task allocator	Assign domain, base version, hypothesis and budget	Prevent overlapping uncontrolled edits
Rule author workers	Inspect traces and propose bounded changes	Many independent candidates
Build and schema service	Compile rules and validate capabilities	Cache by immutable package hash
Simulation workers	Run paired scenarios and stress tests	CPU and memory per active replica
Judge workers	Review selected factual traces	Model tokens, throughput and calibrated panels
Experiment store	Retain manifests, artifacts and metrics	Storage volume and reproducibility
Integration service	Test candidate combinations and gate releases	Interaction testing and reviewer throughput

Independent work and shared definitions

Partition by domain and affected components: care routines, occupation tasks, animal movement, object wear or weather exposure. Shared mechanics such as time, inventory, identity and ownership remain centrally governed contracts. A worker extending them must declare that broader impact so reviewers can run cross-domain scenarios.

Use immutable base snapshots and copy-on-write deltas where practical. Containers isolate candidate plugins and resource use. Workers have access to their task, test data and approved build tools, not permission to edit hidden evaluation sets or production release credentials. This is a correctness boundary: a rule that improves its own grader is not improving the world.

Hundreds or thousands of author agents can be active logically while only a smaller, quota-controlled number of expensive simulations or model calls execute concurrently. The controlling limits are evaluation capacity, inference throughput, storage and merge contention, not just the number of agents with names.

D2 A concrete workshop resource envelope

Separate authoring calls, simulation runs and judge calls. The following envelope is illustrative and independently recalculable. Assume each simulation worker uses two physical-core equivalents and 4 GiB of private working memory for a regional scenario. Shared immutable assets and service overhead are additional.

Concurrent simulation workers	CPU equivalents	Private working RAM	Completed 60-second jobs/hour, ideal
100	200	400 GiB	6,000
1,000	2,000	4,000 GiB	60,000
5,000	10,000	20,000 GiB	300,000

The completion column assumes continuously occupied workers and no queue, loading or retry overhead. It is not a supercomputer benchmark. A “job” is one scenario run; a paired comparison normally consumes at least two runs. If regional complexity doubles memory or duration, capacity changes directly. Never multiply these counts by full-continent memory unless each worker actually needs a continent.

Simulation budget per candidate

For an early gate, 20 scenarios x 10 seeds x baseline/candidate gives 400 runs. At 30 CPU-seconds per run, that costs 12,000 CPU-seconds, or 3.33 core-hours, per candidate. Screening 1,000 candidates costs about 3,333 core-hours before build, inference and integration work. Later long-duration gates should be reserved for survivors.

Model budget per candidate

Suppose four author/repair calls each use 16,000 input and 4,000 output tokens. Add 20 selected episode comparisons, each judged by two independent reviewers using 6,000 input and 1,000 output tokens. The total is 304,000 input and 56,000 output tokens per candidate, or 360 million tokens for 1,000 candidates. At an assumed aggregate output throughput of 10,000 tokens/s, output generation alone takes at least 5,600 seconds. Input processing and round-trip latency add time.

No dollar or GPU count is asserted: those depend on chosen models, provider pricing, batching, context size and deployment. CPU-only world simulation does not imply CPU-only frontier-model authoring. Token budgets belong in the workshop scheduler alongside core-hours and memory.

D3 Evaluation gates and statistical discipline

Use progressively more expensive gates. Mechanical checks run first, then behavioral comparisons, then integration and long-horizon tests. A candidate must earn additional compute by showing a plausible improvement without breaking core constraints.

Gate	Inputs	Pass condition
Static contract check	Schema, units, capabilities, dependency graph	Valid bounded package and migration
Deterministic mechanics	Minimal worlds and explicit expected transitions	No violated invariant or replay mismatch
Paired behavior screen	Baseline/candidate across seeds	Target improvement with acceptable costs
Blind episode review	Same trace format, shuffled identity/order	Evidence-backed rubric improvement
Held-out transfer	New contexts, populations and environments	Improvement extends beyond tuned scenarios
Combined integration	Latest base plus queued candidates	No harmful cross-rule interaction
Long-run and burst test	Days/weeks, crowding, disasters	Stable resources, queues, populations and performance

A rubric with anchors

Score each quality dimension 1-5 with examples: 1 means an unexplained or context-inappropriate result; 3 means a plausible but sometimes repetitive response; 5 means coherent variation with clear causes and good downstream fit. Require cited trace facts and a confidence label. The emotional-intelligence score evaluates represented perception and response, not access to a real person's mind.

Do not average away a hard failure. One duplicated object is not compensated by a high entertainment score. For softer metrics, predeclare non-inferiority margins and trade-offs. Analyze paired differences at the scenario or episode level; individual ticks from one episode are correlated samples. Use confidence intervals and repeated held-out trials, not one preferred story.

Zero observed failures are not proof of safety. Under a simple independent-trial model, zero failures in n trials gives an approximate 95% upper failure-rate bound of $3/n$. Rare situations therefore need targeted construction. Hidden seeds and fresh contexts reduce benchmark overfitting; human calibration and independent judge families reduce shared evaluation bias. These controls must be measured, not assumed successful.

D4 Provisional forks, pull requests and integration

The workshop can evaluate bundles in temporary branches rather than requiring a line-by-line manual review of every parameter. Each candidate must still carry a machine-readable contract and traceable origin. Passing in isolation does not prove that two patches work together.

What a pull request contains

Include base package hash, changed rules and components, intended behavior, dependency versions, schema/migration impact, before/after scenarios, seeds, metrics, judge evidence, resource deltas and a revert plan. Parameter-only changes can use an automated review tier. New schema fields, new queries or engine plugins use a stricter tier with designated owner approval.

How a provisional branch earns release

Integrate compatible candidates into an experiment branch. Replay affected scenarios, cross-domain tests and a fixed representative world suite. Use a dependency graph to select focused tests, but retain a minimum global suite because indirect effects can be missed. If a batch regresses, bisect or test subsets to locate interactions; a good aggregate score cannot excuse a failing subgroup.

Rebase survivors against the latest accepted package and rerun combined checks before merging. GitHub's merge queue is an example of checking changes against an updated target and other queued changes [10]; the architectural requirement is the behavior, not that specific hosting service. Release a pinned manifest only after the complete combination passes.

Limits of parallel improvement

Two rules may be textually independent but behaviorally coupled. A new meal preference can overload restaurant queues. A stricter shift rule can reduce childcare coverage. A more sociable population can explode message traffic. Content merges are cheap compared with discovering these interactions. Track compatibility and performance as first-class results.

Moderators may be agents that prepare decisions, with human owners setting acceptance policy and reviewing difficult trade-offs. Keep authors separate from final graders and release authority. The architecture supports automated low-risk approvals; it does not require unreviewed self-modification of the evaluator or runtime.

D5 Versioned releases and live-world migration

A world pins one rule manifest. A workshop may improve rules continuously while the world continues using its current version. Upgrading requires a declared boundary and a migration plan; it must never rewrite already accepted causes just because a new rule would have chosen differently.

Change	Migration treatment
New numeric default	Applies to new instances; existing values follow an explicit policy
Added need or component	Initialize from declared evidence/defaults; mark newly modeled state
Changed action duration	Finish existing actions under old semantics or migrate progress explicitly
Removed affordance	Resolve active reservations and dependent goals before removal
Changed probabilities	New opportunities use the new policy; old sampled outcomes stay fixed
Terrain generator update	Preserve named geometry and edits; validate routes and locations
New belief rule	Apply on new evidence or a recorded reconsideration event

Stage before broad use

Replay recorded scenarios under the candidate, then run a shadow world or isolated test region. Increase coverage when metrics remain acceptable. A shadow simulation does not mutate production. If versions coexist in neighboring regions, shared protocols and cross-boundary mechanics need an explicit compatibility contract; the MVP should upgrade the whole small world together.

For rule rollback, restore a previous manifest at a future safe boundary and migrate current state as needed. It does not erase events players already observed. Restoring an earlier world checkpoint is a separate branch or explicit administrative repair, with dependent histories considered.

Preserve improvements without losing authorship

Every accepted Beat references the rule manifest and relevant rule IDs. A behavioral regression can therefore be traced to a release, a context or a modeling gap. Keep old schemas and replay readers for retained history. Novel classes and properties become useful only when their semantics, storage cost, migration and evaluation coverage are specified.

The workshop's long-term product is a tested library of reusable behavior, not an ever-growing pile of special cases. Retire redundant rules, merge equivalent definitions and charge complexity against a package budget.

E1 A Python prototype without premature scale claims

Build a transparent reference engine first, with compact storage available from the start. Python owns orchestration, schemas and diagnostics. Critical numeric systems operate on typed arrays. Use a fixed, validated subset of declarative expressions rather than evaluating arbitrary rule text as Python.

Module	Suggested responsibility	Replace later without changing
WorldStore	IDs, component arrays, exact current state	Entity and version contracts
RuleRegistry	Load, validate and pin package manifests	Rule language and capabilities
Scheduler	Timers, next decisions, interrupts	Logical-time semantics
SpatialService	Chunk grid, occupancy, regions, routes	Query and reservation interfaces
ActionResolver	Preconditions, arbitration, effects	Action lifecycle and Beat output
PerspectiveService	Perception, memory and belief updates	Agent evidence restrictions
BeatStore	Append changes, snapshots, replay	Stable event IDs and provenance
Viewer	Pixel sprites, camera, overlays, inspection	Read-only presentation contract
ExperimentRunner	Batch worlds, seeds, metrics and exports	Reproducible experiment manifest

Ordinary dataclasses can describe definitions and small debug worlds. NumPy arrays suit hot state. Numba can compile supported numeric loops [3]; branch-heavy Python object graphs may instead need redesign or a native kernel. Avoid unsafe numeric optimizations when exact replay or threshold decisions depend on floating-point behavior. Fixed-point or quantized values are options where precision permits.

For process parallelism, send compact tasks and use shared immutable assets; serializing an entire world for every tick would waste the benefit. Several independent evaluation worlds are easier to parallelize than one tightly coupled room. Keep a single-process reference mode to compare mechanics with optimized execution.

The viewer should expose time controls, need bars, goals, active rules, reservations, perception coverage and a causal event log. A headless run uses the exact same engine and seeds. Pixel art is an observation tool, not a substitute for tests. This paper specifies that prototype; it does not claim to implement or benchmark it.

E2 Acceptance scenarios and benchmark protocol

Build a small scenario library before inviting large numbers of authors. Every scenario has initial state, allowed stochastic variation, target invariants, metric definitions and an explanation of what a failure means.

Scenario	What must remain true
Urgent need before work	Feasible options; one sampled decision; elapsed-time costs; understandable lateness
Two people, one toilet	No overlapping exclusive use; queue or alternative; no starvation loop
Broken door during travel	Path invalidation; no teleport; perception and repair follow rules
Care obligation vs leisure	Supported context changes priorities without universal stereotypes
Animal and sudden sound	Species-appropriate sensing and bounded response; no remote omniscience
Coarse/fine time-step comparison	Matching rules and comparable event distributions, not per-tick bias
Storyteller promotion and return	Identity, beliefs, action progress and reservations survive
Three fields meet then split	One writer per scope; delivered messages are neither lost nor doubled
Conflicting candidate patches	Integration gate detects combined regression
Week-long neighborhood	No leaked reservations, runaway queues, resource creation or frozen goals

Measure the workloads that will actually ship

Start with 32, then 100, 1,000 and 10,000 actors, holding scenario density and behavior mix explicit. Measure warm-up separately from steady state. Record CPU model, core allocation, RAM, software versions, rules, seeds, world-speed factor and loaded geometry. Benchmark routine background, crowded movement, social interaction, logging and disaster bursts separately, then together.

Report resident memory, peak memory, update distributions, p95/p99 delays, path queries, committed bytes, queue growth and percentage of actors at each fidelity. Repeat on battery and sustained-power modes if laptop play is a target. Tests of one hunger loop cannot validate a million-person city.

At each scale, compare optimized mechanics with the reference engine and compare coarse outcomes with suitable fine runs. Expand only after explaining the bottleneck. The next engineering deliverable should be this neighborhood and benchmark harness, followed by a measured replacement for the capacity assumptions in Appendix C.

References and design status

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[10] GitHub. Managing a merge queue. Official documentation, accessed 27 September 2026.

<https://docs.github.com/en/repositories/configuring-branches-and-merges-in-your-repository/configuring-pull-request-merges/managing-a-merge-queue>

Evidence and scope

This report proposes an architecture, rule interfaces and resource envelopes. It supplies no empirical capacity measurements and no guarantee that automated rule search improves realism. All percentages, costs, schemas and acceptance settings are explicit starting points for implementation and evaluation.

Hardware references establish representative core counts and memory-channel configurations. They do not supply the actor-update costs used in the estimates. The capacity calculations are derived from the stated schema budgets, update rates and world-size assumptions. Library documentation supports implementation options, not claims about a completed prototype.

The behavioral model is intended for authored fictional worlds. Its rules should be tested for internal consistency, believable variation and the intended experience. A claim that it accurately predicts real populations would require separate empirical calibration and validation.